L1.1b Transformations · §1.3

Outside or inside — what it moves, and which way

THE RULE

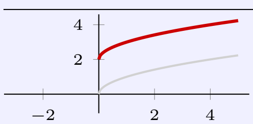
$$\boxed{y = \underbrace{a}_{\mathrm{R}}\, f\Big(\underbrace{b}_{\mathrm{D}}\,(x + \underbrace{c}_{\mathrm{D}})\Big) + \underbrace{d}_{\mathrm{R}}}$$

"Anything outside the $f(x)$ affects RANGE normally. Anything inside the $f(x)$ affects DOMAIN backward."

OUTSIDE | INSIDE

OUTSIDE the $f(x)$
⇒ moves **RANGE**, normally

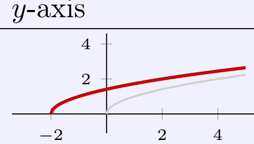
- $f(x) + d$ slide d in the $+y$ direction
- $a\,f(x)$, $|a| > 1$ stretch vertically
- $a\,f(x)$, $0 < |a| < 1$ compress vertically
- $-f(x)$ reflect over the x-axis
- $|f(x)|$ flip whatever is below the x-axis up over it



$\sqrt{x} \;\to\; \sqrt{x} + 2$: R climbs to $[2, \infty)$, D never moves.

INSIDE the $f(x)$
⇒ moves **DOMAIN**, backward

- $f(x + c)$ slide c in the $-\mathbf{x}$ direction †
- $f(bx)$, $|b| > 1$ **compress** horizontally †
- $f(bx)$, $0 < |b| < 1$ **stretch** horizontally
- $f(-x)$ reflect over the y-axis
- $f(|x|)$ chop off the negative x-values and reflect the positive values back over the y-axis

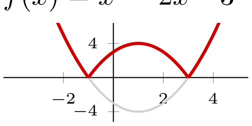


$\sqrt{x} \;\to\; \sqrt{x+2}$: D slides to $[-2, \infty)$ — the $+2$ went the *other* way.

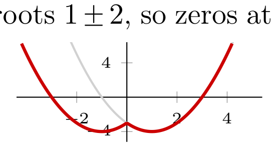
† The two rows that read wrong are the whole point. **Never say "shift left"** — say the direction.

THE ABSOLUTE VALUE PAIR — same bars, opposite sides

$f(x) = x^2 - 2x - 3$ — Loh: $m = 1$, $d = 2$, roots 1 ± 2, so zeros at -1 and 3 and vertex $(1, -4)$.



$|f(x)|$ — **outside**. $R = \{y \mid y \geqslant 0\} = [0, \infty)$



$f(|x|)$ — **inside**. $R = \{y \mid y \geqslant -4\} = [-4, \infty)$

"outside bars bend the outputs up; inside bars chop and mirror the inputs"

FIVE COMBOS — A is f's domain, B is g's

$$\boxed{f \pm g,\quad fg:\quad A \cap B \qquad\qquad \frac{f}{g}:\quad \{\, x \in A \cap B \mid g(x) \neq 0 \,\}}$$

"Intersection cuts out just the overlap of the two domain intervals; union gives you any place that either of them goes."

Complete the square to read a parabola's shifts: $x^2 + 6x + 10 = (x+3)^2 + 1$, vertex $(-3, 1)$ — slide x^2 3 in the $-x$ direction and 1 in the $+y$ direction.

COMPOSITION — diffnifod

$$\boxed{\text{Domain}\big(f(g(x))\big) \;=\; \mathrm{D}_{If} \;\cap\; \mathrm{I}_{f\to\mathrm{O}D}}$$

$$x \to \boxed{g} \to g(x) \to \boxed{f} \to f(g(x))$$

D_{If} $\qquad$ $\mathrm{I}_{f\to\mathrm{O}D}$

D_{If} — the **inner** function has to be defined.

$\mathrm{I}_{f\to\mathrm{O}D}$ — the outer has to be defined too, but its input is $g(x)$: push the outer domain **backward** through the inner.

"Domain of Inner Function, intersect Inner Function to Outer Domain — diffnifod"

Roles, not letters. In $g(f(x))$ the inner function is f, so D_{If} means f's domain.

Four shapes come out: an interval $\left[-\frac{1}{5}, 10\right]$ · a union $(-\infty, -3\,] \cup [\,3, \infty)$ · a hole punched in an interval $[\,0, 4) \cup (4, \infty)$ · **empty** $\varnothing$.

WATCH OUT

✗ ~~$f(x+2)$ shifts the graph 2 to the right~~ → **inside, so backward — slide 2 in the $-x$ direction**

✗ ~~$f(3x)$ stretches it horizontally by 3~~ → **inside, so backward — $|b| > 1$ *compresses* by 3**

✗ ~~$|f(x)|$ and $f(|x|)$ do the same thing~~ → **outside bars change outputs, inside bars change inputs**

✗ ~~simplify first, then read the domain off the answer~~ → $(\sqrt{x})^2 - 9 = x - 9$ **looks defined everywhere — the domain still remembers** $[0, \infty)$

✗ ~~the composite's domain is the outer function's domain~~ → **intersect both halves —** D_{If} ***and*** $\mathrm{I}_{f\to\mathrm{O}D}$

✗ ~~every composite has a domain~~ → $\sqrt{-9 - x^2}$ **— no overlap, nothing to cut out, so** $\varnothing$